



MAKHTESHIM
CHEMICAL WORKS LTD.

MATERIAL SAFETY DATA SHEET

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Revised edition no : 4

Date : 2 / 8 / 2011

Supersedes : 16 / 11 / 2009

MIRAGE 45 EC(NA)

R-7168.DPD

1. Identification of the substance/mixture and of the company/undertaking

Trade name : MIRAGE 45 EC(NA)
Product code : R-7168.DPD
Type of product : Fungicide
Chemical formula : C₁₅H₁₆Cl₃N₃O₂
Company identification : Makhteshim Chemical Works Ltd.
PO Box 60
Beer Sheva 84100 Israel.
Fax : +972-8-6296848
MSDS@ma-industries.com
Emergency phone nr : 972-3-6106666 Subscription no. 36789 (Makhteshim)
Email : MSDS@ma-industries.com

2. Hazards identification

Environmental hazard : Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

3. Composition/information on ingredients

This product is considered to be hazardous and contains hazardous components.

Substance name	Value(s)	CAS No	EC No	Index No	REACH Reg.	Classification
Prochloraz	: 450 g/l	67747-09-5	266-994-5	613-128-00-2	-----	Xn; R22 N; R50-53
Butyrolactone	: ca 4 %	96-48-0	202-509-5	-----	-----	Xn; R22 R67 Xi; R41
n-Butyl alcohol	: ca 4 %	71-36-3	200-751-6	603-004-00-6	-----	R10 Xn; R22 R67 Xi; R37/38-41

4. First aid measures

Effects and symptoms
Eye contact : Redness , tears .
First aid
Ingestion : Do not induce vomiting. Never give anything by mouth to an unconscious person.
Wash out mouth with plenty of water. Consult a doctor.
Skin contact : Remove affected clothing and wash all exposed skin area with soap and water,
followed by warm water rinse. Consult a doctor in the event of any complaints.
Eye contact : If pain persists, consult an eye specialist.
Wash off with plenty of water
Inhalation : Remove victim to fresh air. Consult a doctor in the event of any complaints.
Note to physician : No specific antidote known.
Treat symptomatically and give supportive therapy.
If ingested perform gastric lavage and administer activated charcoal.

5. Fire-fighting measures

Extinguishing media : If involved in fire: water spray , water fog , foam .

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5. Fire-fighting measures (continued)

- Special exposure hazards** : Thermal decomposition may generate : carbon dioxide , carbon monoxide , chlorides , nitrogen oxides .
- Protection against fire** : Wear suitable protective clothing Self-contained breathing apparatus.
- Special procedures** : Fight fires from a protected location . Dike fire control water for later disposal.

6. Accidental release measures

- Personal precautions** : Use appropriate protection (see section 8).
- Environmental precautions** : Dispose of this material and its container at hazardous or special waste collection point In accordance with national and regional regulations . If the product has contaminated surface water, inform the appropriate authorities. Contaminated soil layers have to be dug out.
- After spillage and/or leakage** : In the event of minor spillage: Absorb in sand or other inert material . Use appropriate container to avoid environmental contamination.
In the event of major spillage: Collect and contain as much free liquid as possible. Dike spills using absorbent or impervious materials such as sand or clay for later disposal.

7. Handling and storage

- Handling** : Do not breathe fumes . Avoid contact with skin and eyes.
- Storage** : Keep locked up . Keep in original containers. Keep container tightly closed and in a well-ventilated place (<50°C).

8. Exposure controls/personal protection

- Industrial hygiene** : Facilities storing or utilizing this material should be equipped with an eyewash facility and safety shower . Wash hands thoroughly after handling. Wash clothing before re-using.
- Personal protection**
- **Respiratory protection** : During spraying wear suitable respiratory equipment.
 - **Skin protection** : Wear protective clothing and chemical resistant boots .
 - **Hand protection** : Wear suitable gloves.
 - **Eye protection** : Chemical goggles or safety glasses.
- Occupational Exposure Limits**
- TLV** : Prochloraz : Not established

9. Physical and chemical properties

- Physical state** : Liquid
- Colour** : Clear amber solution
- Odour** : characteristic
- Specific gravity** : (H₂O = 1) 1.13 - 1.14
- Molecular weight** : Prochloraz : 376.7

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9. Physical and chemical properties (continued)

Melting point [°C]	: Prochloraz : 48 - 51 (Pure)
Vapour pressure	: Prochloraz : 4.5X (10) ⁻⁶ Pa at a temperature of 25°C (Pure)
Solubility in water	: Emulsifiable Prochloraz : 25 ppm at a temperature of 20°C (Pure)
Flash point [°C]	: 66 (Closed cup)
Log P octanol	: Prochloraz : log Pow = 4.06
Explosion limits	: Not explosive

10. Stability and reactivity

Physico-chemical stability	: Stable under normal conditions.
Hazardous decomposition products	: Thermal decomposition may generate : carbon dioxide , carbon monoxide , nitrogen oxides , chlorides .
Hazardous reactions	: Hazardous polymerization will not occur .
Materials to avoid	: Avoid contact with : strong acids , strong bases .
Conditions to avoid	: Protect from sunlight , open flame , sources of heat . Decomposes upon heating.

11. Toxicological information

Oral, Rat LD50 [mg/kg]	: > 2000
Dermal, Rabbit LD50 [mg/kg]	: > 2000
Inhalation, Rat LC50 [mg/l/4h]	: > 4.99
Skin corrosion/ irritation	: Mildly irritating to skin
Serious eye damage/ irritation	: Mildly irritating to eyes
Respiratory or skin sensitization	: Guinea pig maximization test: Non sensitizer
Carcinogenicity	: Prochloraz : Not carcinogenic (rat)
Germ cell mutagenicity	: Prochloraz : Not mutagenic
Reproductive toxicity	: Prochloraz : Not teratogenic in animal experiments

12. Ecological information

96 H-LC50 - Rainbow trout [mg/l]	: 3.05
48 Hour-EC50 - Daphnia magna [mg/l]	: 4.00
72 H-EbC50 - Algae [mg/l]	: 1.42
72 H-ErC50 - Algae [mg/l]	: 4.34
LD50 Birds [mg/kg]	: Prochloraz Technical: > 2000
Bees LD50 [µg/Bee]	: Prochloraz Technical: Not toxic to bees
Persistence and degradability	: Rapidly degrades when exposed to sunlight and UV light .
Mobility	: Product does not leach to ground water.

13. Disposal considerations

Disposal	: Dispose in a safe manner in accordance with local/national regulations.
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13. Disposal considerations (continued)

14. Transport information

- UN No. : 3082
- Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (450 g/l PROCHLORAZ)
- Packing group : III
- Class : 9
- H.I. nr : 90
- IMDG-Marine pollution : YES

15. Regulatory information

EU Classification : This product is provisionally labelled by the supplier in accordance to the EU regulations:

Symbol(s)



R Phrase(s)

- : N: Dangerous for the environment.
- : R51/53 - Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

S Phrase(s)

- : S02 - Keep out of the reach of children.
- : S13 - Keep away from food, drink and animal feedingstuffs.
- : S20/21 - When using, do not eat, drink or smoke.
- : S61 - Avoid release to the environment. Refer to special instructions/Safety data sheets.

16. Other information

Recommended uses and restrictions : None.

List of full text of R-phrases in section 3.

- : R10 - Flammable.
- : R22 - Harmful if swallowed.
- : R37/38 - Irritating to respiratory system and skin.
- : R41 - Risk of serious damage to eyes.
- : R50/53 - Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.
- : R67 - Vapours may cause drowsiness and dizziness.

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